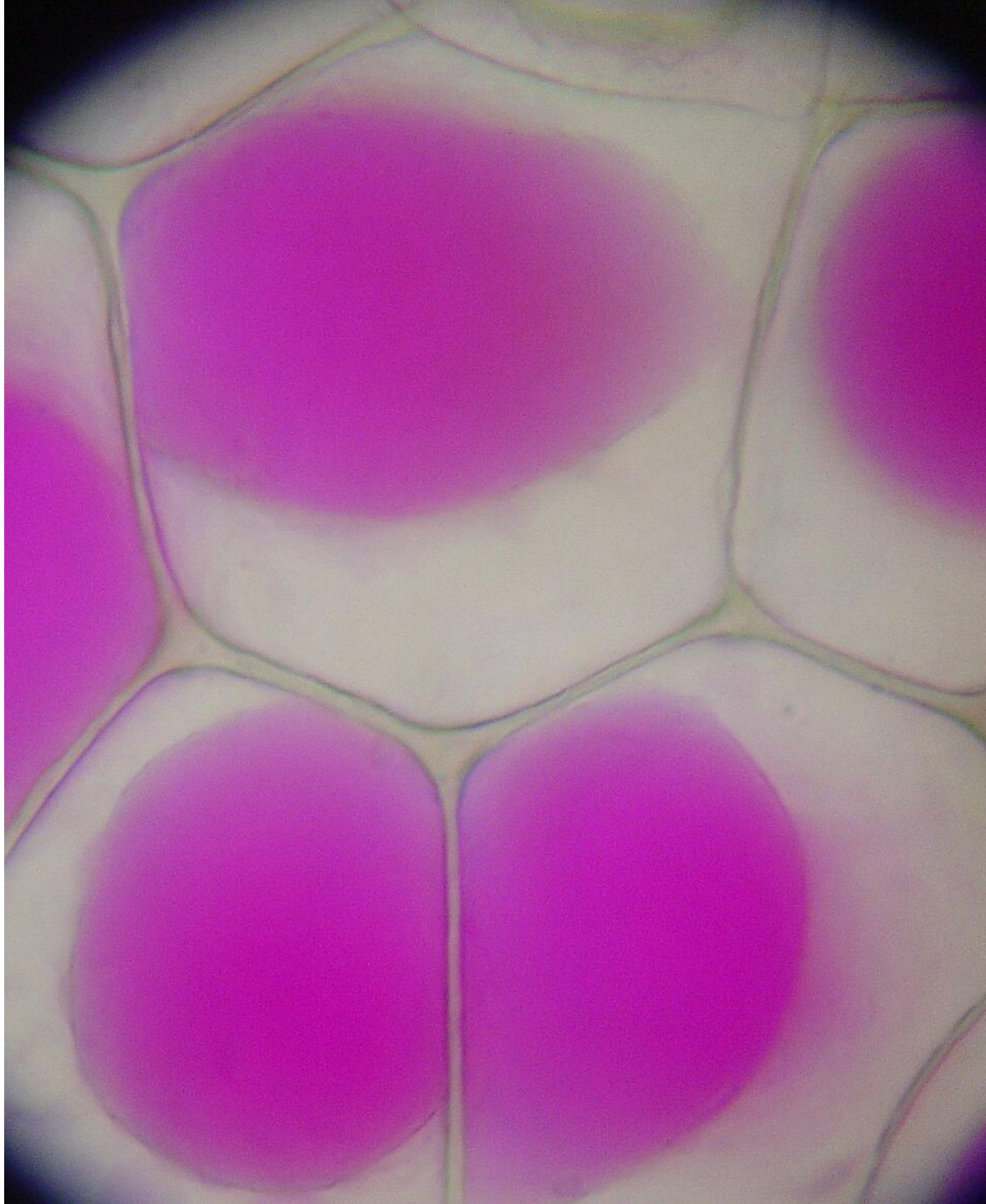
DP Biology – D2.3

Water potential

- What affects whether water moves into or out of a cell?
- In what ways do plant cells and animal cells control water movement differently?

SL statements D2.3.1–7. Textbook pp. 684–690.
Image: plasmolysed plant cells. CC BY-SA 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D2.3.1–4	Explain solvation and osmosis, use the terms hypotonic, hypertonic and isotonic, and analyse data from plant tissue
2	D2.3.5–7	Explain what water movement does to cells with and without walls, and why isotonic solutions are used in medicine

Why water matters to cells

D2.3.1–7

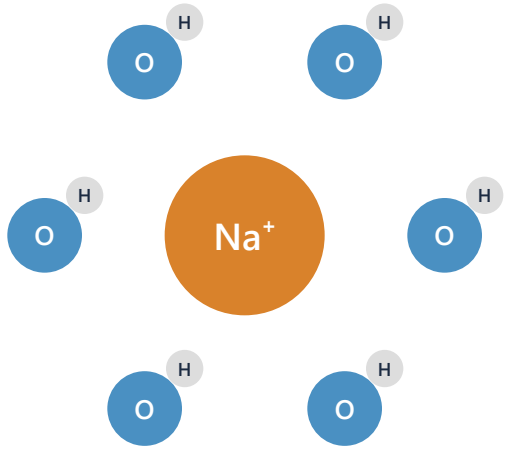


Roses wilting after a heat wave. W. Carter, CC0.

These roses wilted because their cells lost water. Give them water and they stand up again.

In this chapter: why water moves into or out of cells, and why plant cells survive changes that would burst an animal cell.

Water as a solvent



water molecules surround the ion

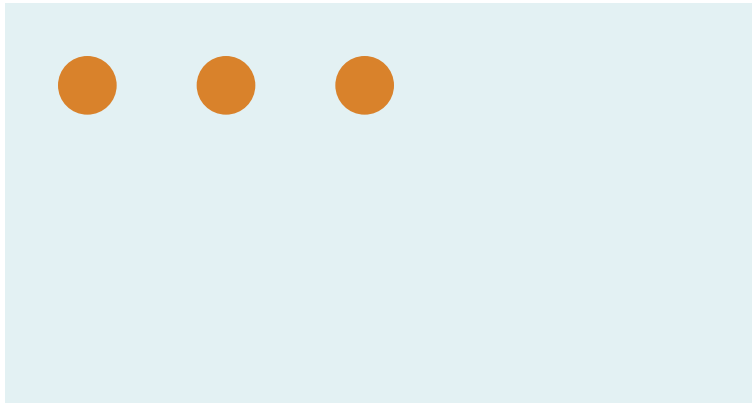
Water molecules are **polar**: the oxygen end is slightly negative, the hydrogen end slightly positive.

They cluster round ions and polar molecules, pulling them apart and keeping them in solution: **solvation**.

So salts, sugars and amino acids dissolve in the cytoplasm, ready to react.

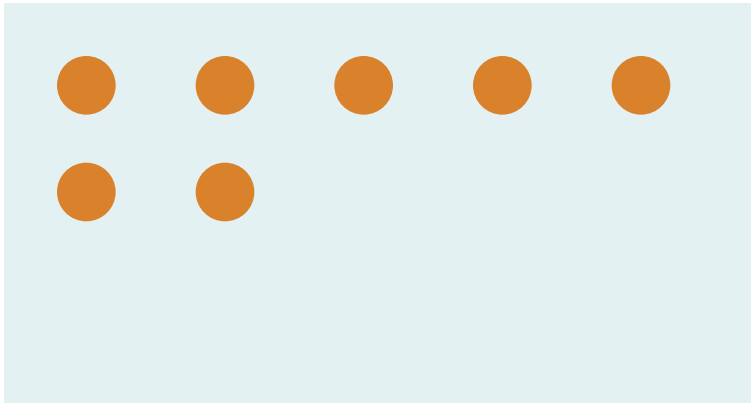
Comparing solutions

D2.3.2



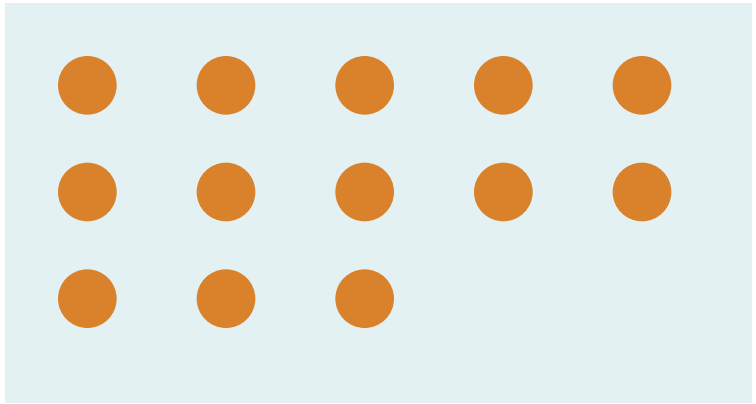
Hypotonic

lower solute concentration



Isotonic

same solute concentration



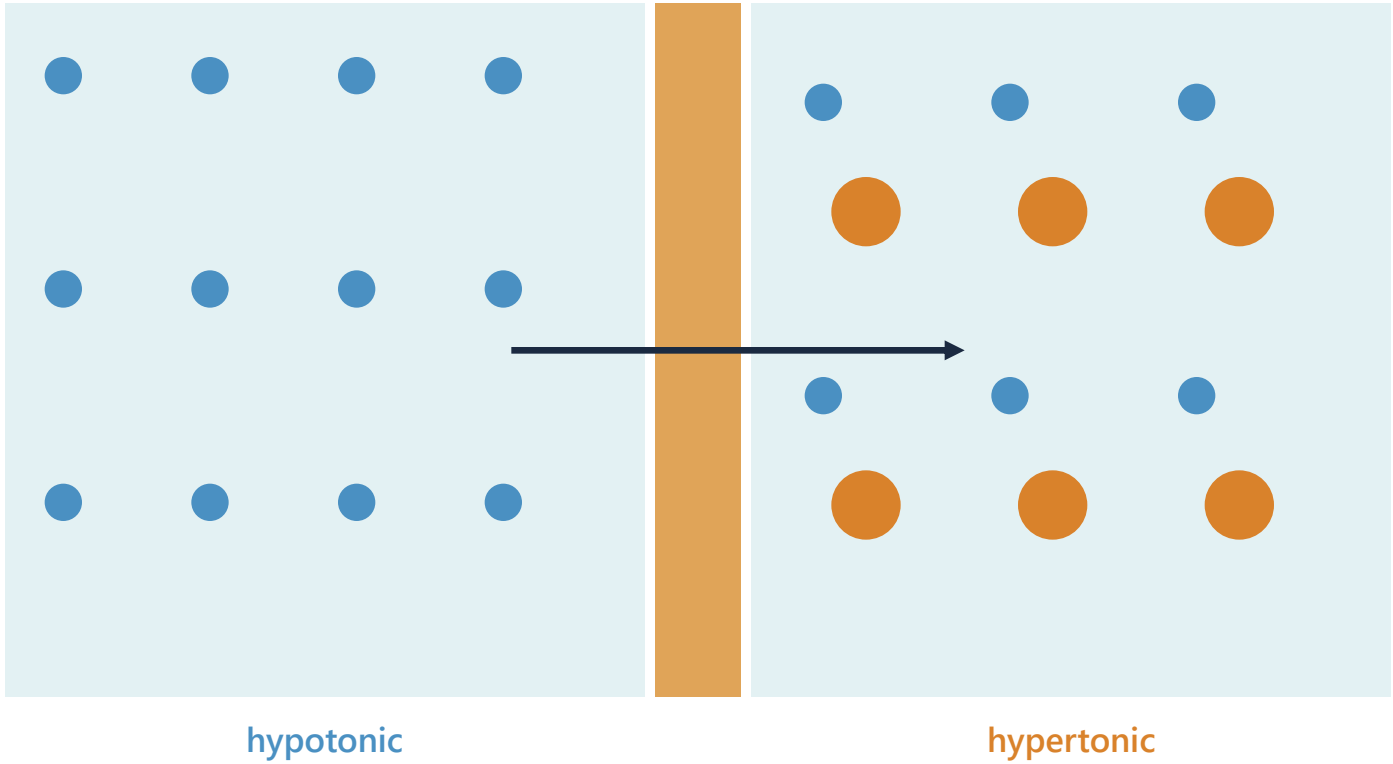
Hypertonic

higher solute concentration

Always compare two

These words compare one solution with another, usually with the inside of a cell. A solution is hypertonic only compared with something else.

Osmosis: water follows solute

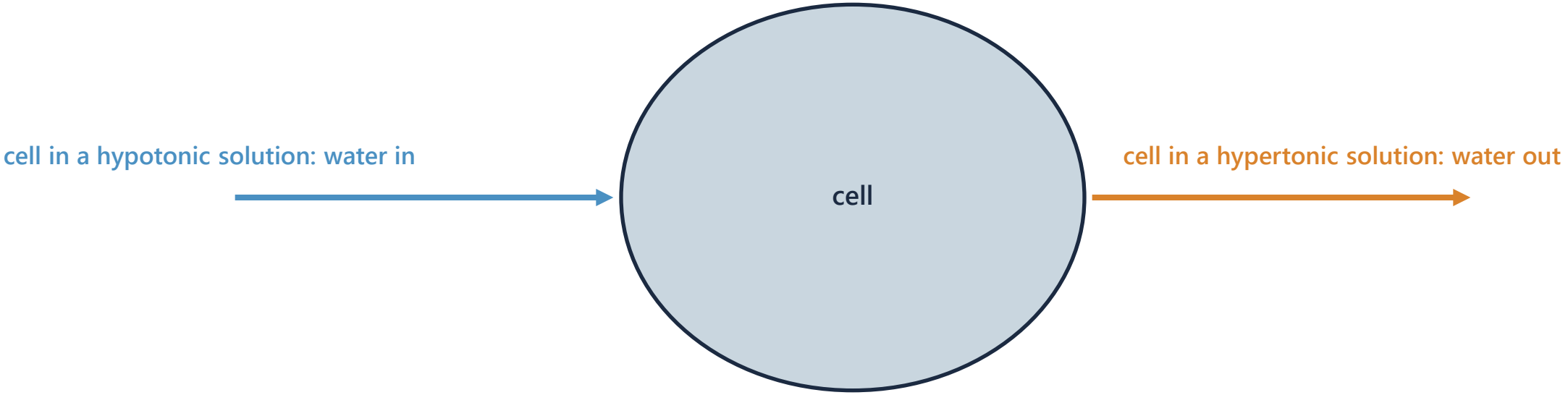


Water moves across a membrane from the **lower** solute concentration to the **higher** one.

The solute cannot cross, so water moves instead. This is **osmosis**, and it needs no energy.

Water entering and leaving cells

D2.3.3

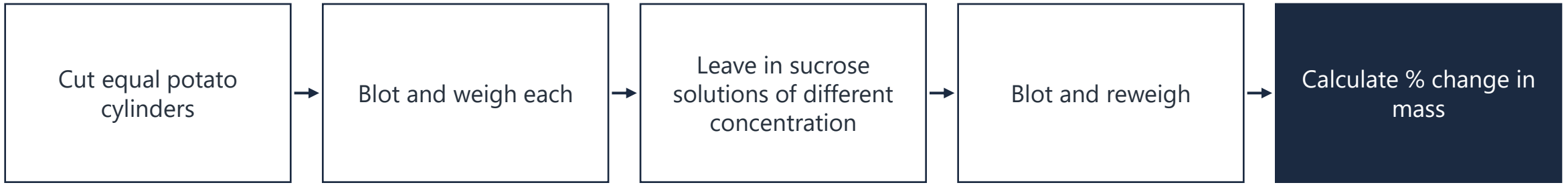


Key idea

Water moves towards whichever side has the higher solute concentration. In an isotonic solution, water moves both ways equally: no net change.

Practical: potato in sucrose

D2.3.4

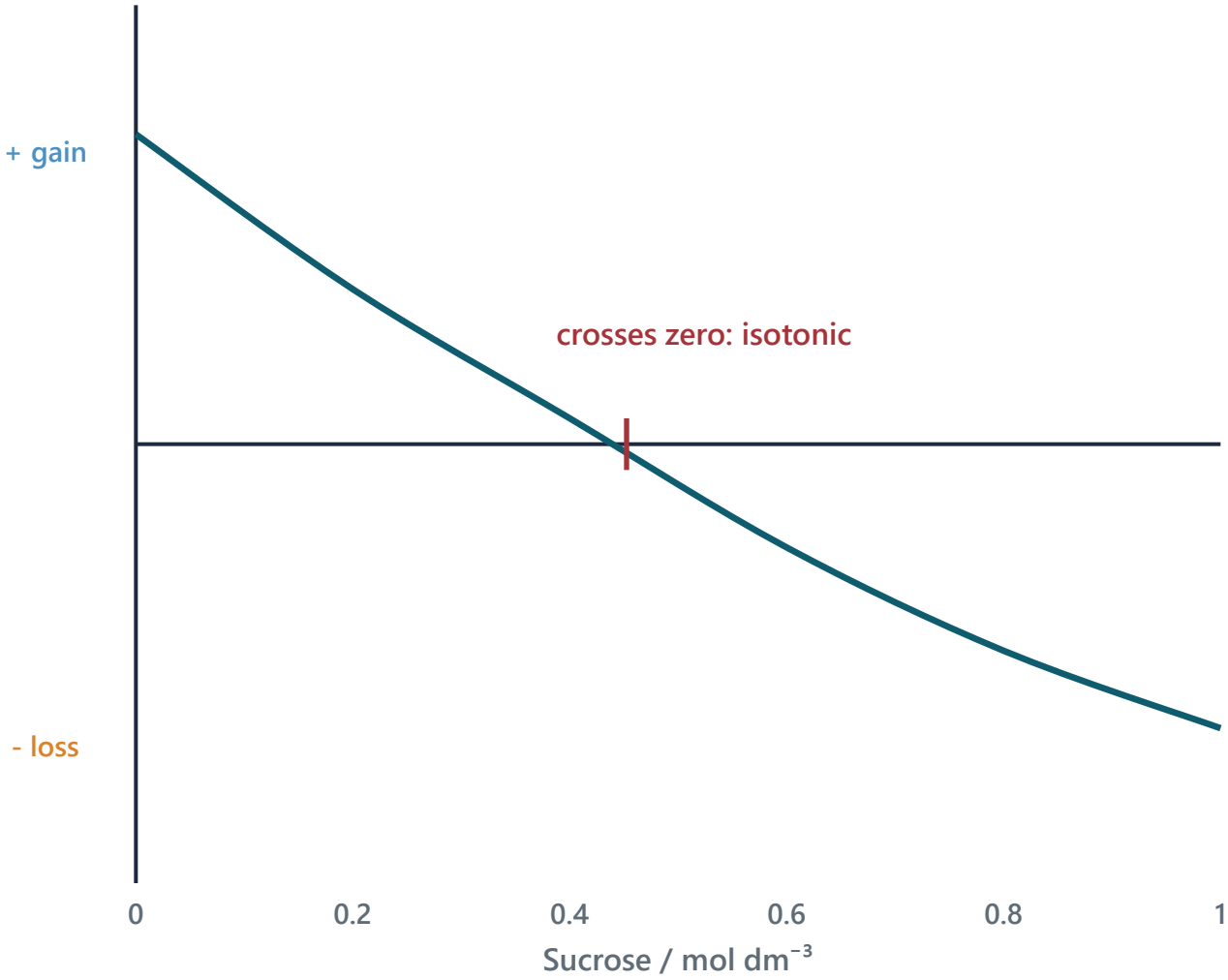


$$\% \text{ change} = (\text{final mass} - \text{initial mass}) / \text{initial mass} \times 100$$

Use percentage change because the cylinders start at slightly different masses.

Gain in mass: the solution was hypotonic to the potato. Loss in mass: hypertonic. No change: isotonic.

Reading osmosis data



Plot % change against concentration.

Where the line crosses **zero**, there is no net water movement: that solution is **isotonic** with the potato cells.

Read it off the x-axis to estimate the solute concentration of the cells.

Working with data: potato cylinders

D2.3.4

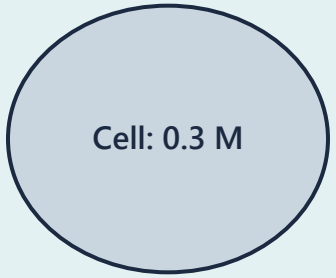
Mean change in mass of potato cylinders after 24 hours. These data are invented for practice; the pattern is realistic.

Sucrose / mol dm^{-3}	Start / g	End / g	% change
0.0	2.50	2.80	?
0.2	2.40	2.52	?
0.4	2.60	2.55	?
0.6	2.50	2.30	?

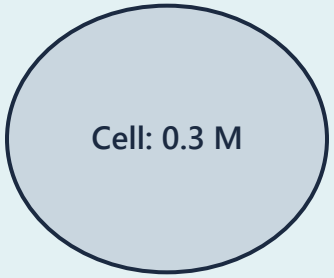
1. Calculate each % change.
2. Estimate the isotonic concentration.
3. Explain the result at 0.6.

Practice: predict the direction

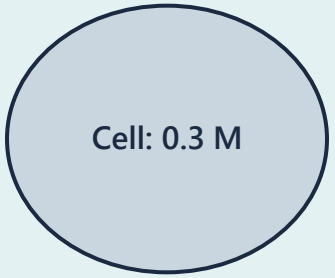
D2.3.2–3



Outside: 0.1 M



Outside: 0.3 M



Outside: 0.6 M

1. For each, name the outside solution as hypotonic, isotonic or hypertonic.
2. State the net direction of water movement.

Check-in: class 1

D2.3.1–4

1. Explain why salts dissolve in water.
2. Define hypotonic, isotonic and hypertonic.
3. Define osmosis.
4. Explain why % change in mass is used in the potato practical.
5. Explain how to find the isotonic concentration from a graph.

How to use this

Answer without your notes, then check against slides 4–11.

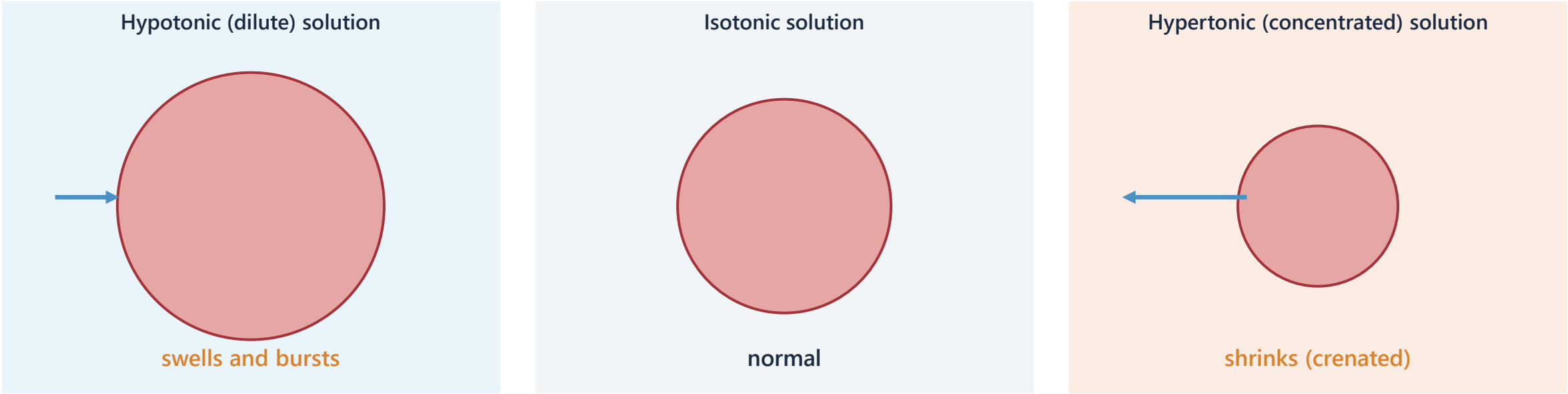
Water and solute: say it right

D2.3.2–3

Not this	But this
Water moves from high to low concentration.	Water moves from low to high solute concentration.
The solute moves by osmosis.	Only water moves by osmosis.
Hypertonic means concentrated.	Hypertonic means more concentrated than another solution.

Cells without a cell wall

D2.3.5

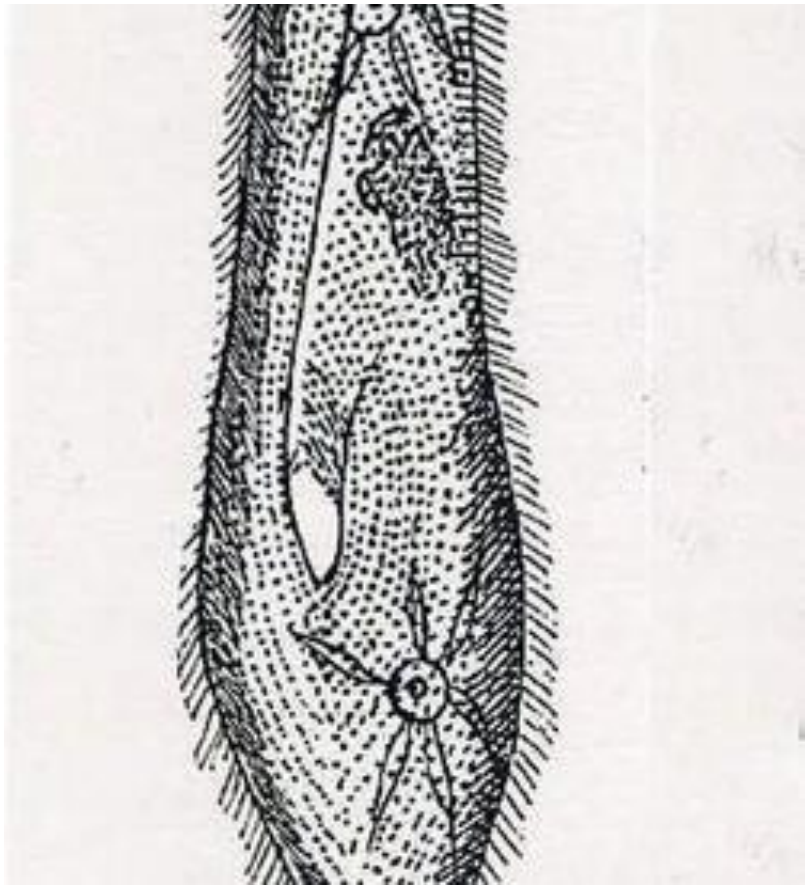


Animal cells have no wall. In a hypotonic solution water floods in until the membrane **bursts**. In a hypertonic solution water leaves and the cell **shrinks**.

So animals must keep their body fluids **isotonic** with their cells.

Single cells in fresh water

D2.3.5



Paramecium. Public domain.

Pond water is hypotonic to Paramecium, so water keeps entering.

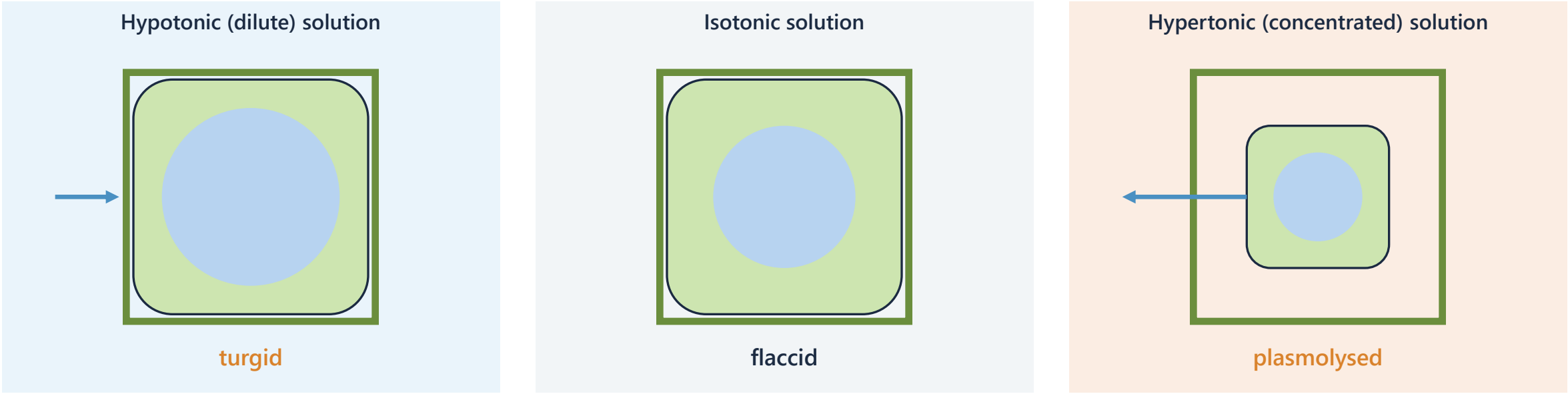
It survives by using **contractile vacuoles** that fill with water and pump it out, using energy.

Think

What would happen if the contractile vacuoles stopped working?

Cells with a cell wall

D2.3.6



In a hypotonic solution the vacuole swells and pushes against the strong **wall**: the cell becomes **turgid** but does not burst.

In a hypertonic solution water leaves and the membrane pulls away from the wall: **plasmolysis**.

Turgor holds plants up

D2.3.6



Wilting roses. CC0.

Turgid cells press against each other and keep soft stems and leaves firm.

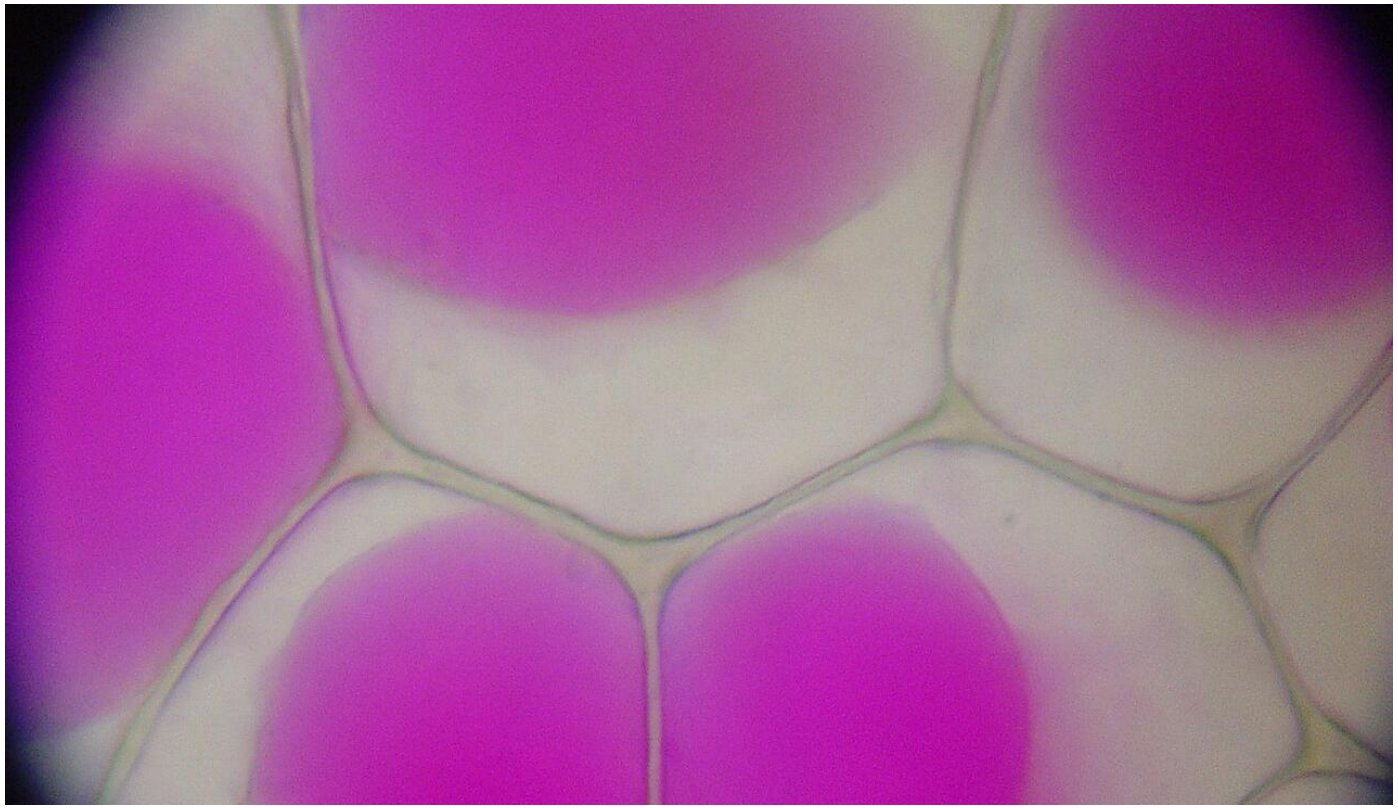
When water is short, cells become **flaccid** and the plant **wilts**.

Link

This is why plants need a steady water supply from the roots (B3.2 transport in plants).

Seeing plasmolysis

D2.3.6



Plasmolysed Rhoeo cells: coloured contents pulled away from the walls. CC BY-SA 3.0.

In salt solution, the coloured cytoplasm and vacuole shrink away from the cell wall.

The wall keeps its shape: the gap fills with the outside solution.

Try it

Mount red onion skin in water, then in salt solution. Watch the colour shrink from the corners.

Plant and animal cells compared

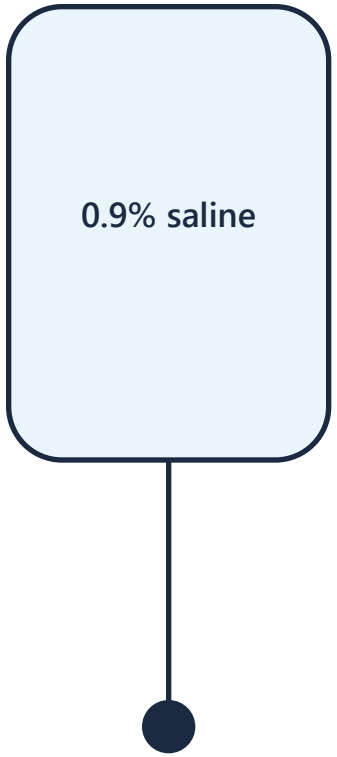
D2.3.5–6

Solution	Animal cell	Plant cell
Hypotonic	swells and bursts	turgid; wall stops it bursting
Isotonic	no change	flaccid
Hypertonic	shrinks	plasmolysed

Why the difference?

Only the plant cell has a strong cellulose wall that pushes back as the cell swells.

Isotonic solutions in medicine



Fluids given into a vein use **isotonic** saline (0.9% salt), so blood cells neither burst nor shrink.

Donor organs for transplant are packed in cold isotonic solution, so their cells keep their shape.

Eye drops and contact lens solution are isotonic too, so they do not sting.

Working with data: red blood cells in salt

D2.3.5, 7

Percentage of red blood cells that burst after 10 minutes. These data are invented for practice; the pattern matches real results.

Salt / %	Burst / %
0.0	100
0.3	85
0.5	20
0.7	2
0.9	0

1. Describe the trend.
2. Explain why cells burst in low salt.
3. Suggest why 0.9% salt is used in drips.
4. Predict what cells look like in 3% salt.

Practice: explain what happens

D2.3.5–7

Salting slugs

Salt on a slug’s moist skin.

Fresh fish in the sea

A freshwater fish is put in sea water.

Too much fertiliser

Strong fertiliser around a plant’s roots.

How to use this

Use the words hypertonic, osmosis and water potential (or solute concentration) in each answer.

Check-in: class 2

D2.3.5–7

1. Describe what happens to a red blood cell in pure water.
2. Explain why a plant cell does not burst in pure water.
3. Define plasmolysis.
4. Explain how Paramecium survives in fresh water.
5. Explain why drips use isotonic saline.

How to use this

Answer without your notes, then check against slides 14–22.

Check that you can...

1. **Explain** solvation
2. **Define** hypotonic, isotonic and hypertonic
3. **Explain** osmosis in terms of solute concentration
4. **Calculate** % change in mass and **analyse** potato data
5. **Compare** what happens to animal and plant cells in different solutions
6. **Explain** why isotonic solutions are used in medicine

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Define hypertonic.
2. Explain what happens to a plant cell in a hypertonic solution.
3. Explain why isotonic saline is used in medical drips.

Image credits

Rhoeo Discolor – Plasmolysis.jpg – Mnolf – CC BY-SA 3.0

Roses wilted after heat wave at Gamla Strandgatan 11, Gamlestan, Lysekil.jpg – W.carter – CC0

Paramecium caudatum.jpg – Alfred Kahl – Public domain

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses